

Generative Paper 6: Elementary Fiber Dynamics — The Building Block of Photon Formation

Author Information

Author: Mortaza Zabihi

Affiliation: Independent Researcher

Email: m.zabihi.research@gmail.com

Date: July 2026

Version: 1.0

Preprint — first posted on Zenodo, DOI: <https://doi.org/10.5281/zenodo.21718571>

Abstract

This paper introduces the elementary organizational interaction underlying the ψ - θ - χ framework by isolating the smallest possible dynamic process occurring within a single fiber.

Rather than beginning from particles, fields, or spacetime geometry, the framework assumes two continuous organizational fields, ψ and θ , whose temporary interaction gives rise to a discrete organizational transition denoted by χ . The appearance of χ does not represent the motion of an independent entity but the transient organizational bridge through which local incompatibility between ψ and θ is temporarily resolved.

The resulting dynamics are intrinsically hybrid. While ψ and θ evolve continuously, χ exists only as a discrete event associated with coupling and decoupling. The elementary interaction is therefore described as an event-driven organizational process rather than a continuously evolving geometric structure.

The purpose of this paper is deliberately limited. It establishes the minimal organizational event from which all higher-order processes of the ψ - θ - χ framework are constructed. No assumptions are made regarding propagation, collective organization, photons, or electromagnetic phenomena. Those organizational levels require additional principles that lie beyond the scope of a single elementary interaction.

By isolating the simplest organizational dynamics, this paper provides the conceptual and mathematical foundation upon which the subsequent development of photon formation and collective electromagnetic organization is built.

Keywords: Photon; Organizational Fields; Generative Architecture; Generative Framework;

© 2026 Mortaza Zabihi. Distributed under Creative Commons Attribution (CC BY 4.0) License.

1. Introduction

Modern theoretical physics successfully describes the behavior of electromagnetic fields and elementary particles through sophisticated mathematical formalisms. Nevertheless, these descriptions generally begin with entities that are already assumed to exist. Fields, particles, and spacetime provide the starting point, while the organizational mechanisms responsible for their emergence remain outside the scope of the theory.

The present work adopts the opposite perspective.

Instead of asking how existing physical entities evolve, it asks what elementary organizational process could exist prior to any stable physical structure.

Within the ψ - Θ - χ framework, every higher-order phenomenon is assumed to originate from the repeated reconstruction of a single elementary interaction. Understanding this elementary process is therefore a prerequisite for understanding every subsequent organizational level.

For this reason, the analysis deliberately isolates one fiber and one local interaction.

The objective is not to construct a realistic model of electromagnetic propagation, nor to explain photons, collective organization, or observable wave phenomena. Such structures require organizational mechanisms that cannot exist within an isolated elementary interaction.

The purpose of this paper is considerably more modest.

It seeks to identify the smallest organizational event capable of producing a transient but internally consistent interaction between two continuous organizational fields.

To accomplish this, the framework introduces two continuous organizational variables, ψ and Θ , together with a discrete organizational transition χ that temporarily bridges their local incompatibility during coupling. The coexistence of continuous and discrete organizational processes naturally gives rise to a hybrid dynamical system whose behavior cannot be reduced to either purely continuous or purely discrete evolution.

Throughout this paper, organization is treated as fundamental, while geometry is intentionally excluded from the elementary description. No assumptions are made regarding spatial neighborhoods, propagation paths, or collective structures. The elementary interaction is analyzed solely as a local organizational event whose reconstruction will later serve as the building block for photon formation and, subsequently, collective electromagnetic organization.

The remainder of the paper develops this elementary interaction in a progressive manner. Section 2 introduces the organizational architecture of ψ , Θ , and χ . Section 3 analyzes the coupling and decoupling dynamics governing a single interaction cycle. Section 4 formulates the resulting hybrid dynamical system. Finally, Section 5 clarifies the scope and limitations of the present formulation and outlines its role as the organizational foundation for the subsequent study of photon formation.

2. Organizational Architecture of the Elementary Interaction

The ψ - θ - χ framework begins with the simplest possible organizational system capable of producing a transient interaction.

At this level, no assumptions are made concerning photons, propagation, collective behavior, or electromagnetic organization.

Only one elementary interaction is considered.

The objective is to identify the minimal organizational architecture required for such an interaction to exist.

This architecture consists of two continuous organizational fields, ψ and θ , together with a discrete organizational transition χ .

Each component possesses a distinct organizational role.

2.1 The Continuous Organizational Fields

The framework assumes two continuously existing organizational fields, denoted by ψ and θ .

These fields should not be interpreted as electromagnetic fields, quantum wavefunctions, or physical media.

They represent continuous organizational states whose evolution is independent of any particular physical interpretation.

Their continuity implies that organizational change occurs without intrinsic discontinuities.

Neither ψ nor θ appears or disappears during ordinary evolution.

Instead, both continuously modify their local organizational states.

Because the two fields evolve independently, situations inevitably arise in which their local organizational states become temporarily incompatible.

This incompatibility does not imply physical collision or geometric overlap.

It simply indicates that the simultaneous continuation of the two local organizational states cannot be organizationally maintained without a transient restructuring process.

The purpose of χ is precisely to provide this temporary restructuring.

2.2 Organizational Stress

The temporary incompatibility between ψ and θ is described as **organizational stress**.

Stress is not introduced as a mechanical force, stored energy, or physical pressure.

Instead, it measures the inability of the current local organizational states of ψ and θ to remain simultaneously consistent.

As long as the organizational states remain mutually compatible, no additional organizational process is required.

When incompatibility exceeds the admissible organizational limit, the existing configuration can no longer persist continuously.

A discrete organizational transition therefore becomes necessary.

Stress should therefore be understood as the organizational condition that enables the appearance of χ rather than as an independently propagating physical quantity.

2.3 The Discrete Organizational Transition χ

Unlike ψ and θ , χ is not continuously present.

It exists only during the temporary organizational restructuring required to resolve local incompatibility.

The appearance of χ therefore marks the occurrence of an elementary organizational event.

Equally important, χ is not an independent physical object.

Its organizational function is to establish a temporary bridge through which ψ and θ become locally coupled.

Once the coupling has completed its elementary organizational reconstruction, χ disappears.

The lifetime of χ is therefore intrinsically finite.

Every elementary interaction begins with the appearance of χ and terminates with its disappearance.

The organizational dynamics are consequently event-driven rather than continuously mediated.

2.4 χ as an Elementary Organizational Bridge

The organizational role of χ extends beyond merely indicating that a coupling event has occurred.

During its finite lifetime, χ establishes the temporary organizational bridge through which the local restructuring of ψ and θ becomes possible.

This bridge possesses no independent persistence.

It cannot exist without the organizational states that generate it.

Likewise, it cannot survive after the elementary reconstruction has been completed.

The elementary interaction is therefore neither a direct interaction between ψ and θ nor an autonomous evolution of χ .

Instead, it is a transient organizational process whose existence depends simultaneously upon all three organizational components.

This interpretation distinguishes χ from conventional mediating particles or continuously existing interaction fields.

χ is neither transmitted nor conserved.

It is created, performs its organizational role, and disappears.

2.5 The Minimal Organizational Architecture

The elementary ψ - θ - χ interaction establishes the smallest organizational structure considered within the present framework.

Only three assumptions are required:

- ψ and θ exist as continuous organizational fields.
- Temporary organizational incompatibility may arise locally between them.
- χ appears only as a transient organizational bridge that enables local reconstruction.

No assumptions concerning spatial geometry, propagation, neighboring fibers, collective organization, or persistent physical structures are introduced at this level.

Those phenomena require organizational mechanisms that emerge only after many elementary interactions become recursively organized.

The present paper therefore isolates the elementary organizational architecture upon which every subsequent level of the ψ - θ - χ framework will be constructed.

3. Elementary Coupling Dynamics

The previous section established the organizational architecture of the elementary ψ - θ - χ interaction.

The present section examines how this architecture evolves through one complete elementary organizational event.

At this level, the dynamics remain entirely local.

No propagation occurs.

No organizational memory is assumed.

No interaction extends beyond the single elementary event under consideration.

The objective is to identify the minimal sequence through which two continuous organizational fields temporarily couple and subsequently separate.

3.1 Coupling Initiation

An elementary interaction begins when the local organizational states of ψ and θ become temporarily incompatible.

As discussed previously, this incompatibility is expressed as organizational stress.

The appearance of stress alone does not constitute an interaction.

It merely establishes the organizational condition under which a discrete transition may occur.

When the local organizational state can no longer evolve continuously, χ is generated.

The emergence of χ marks the initiation of coupling.

At this instant, ψ and θ remain continuous organizational fields.

Only χ appears discretely.

The interaction therefore begins with the coexistence of continuous organizational evolution and a newly generated discrete organizational bridge.

3.2 The Coupled Organizational State

Once χ has appeared, ψ and θ become temporarily connected through the elementary organizational bridge established by χ .

The coupled state is neither permanent nor self-sustaining.

Its existence depends entirely upon the continued presence of χ .

During this interval, local organizational stress is progressively reorganized through the temporary coupling.

No physical quantity is transferred.

No permanent structure is created.

The coupled state should therefore be understood as a transient organizational configuration rather than a stable object.

Its purpose is solely to enable local organizational reconstruction.

3.3 Decoupling and the Disappearance of χ

The elementary interaction concludes when the temporary organizational reconstruction has been completed.

At this point, the organizational role of χ no longer exists.

The bridge disappears.

The coupled state terminates.

ψ and θ continue their independent continuous evolution.

Nothing persists from χ itself.

Its existence is strictly limited to the finite interval during which coupling remains organizationally necessary.

The disappearance of χ therefore does not represent the destruction of an object.

It simply marks the completion of one elementary organizational event.

3.4 The Elementary Interaction Cycle

The complete elementary dynamics may now be summarized as a four-stage organizational sequence:

1. Independent continuous evolution of ψ and θ .
2. Local organizational incompatibility giving rise to stress.
3. Temporary appearance of χ and formation of the coupled organizational state.
4. Disappearance of χ and restoration of independent continuous evolution.

This sequence defines the smallest organizational cycle considered within the $\psi-\theta-\chi$ framework.

Importantly, the cycle itself does not imply periodic repetition.

The present analysis describes only one elementary event.

Whether such events can become recursively organized into persistent cyclic structures remains outside the scope of this paper.

3.5 Memoryless Organizational Dynamics

An essential property of the elementary interaction is the absence of organizational memory.

Each coupling event is determined entirely by the instantaneous local organizational states of ψ and θ .

The appearance of χ depends only on the present organizational configuration.

Likewise, the disappearance of χ leaves no persistent organizational bridge connecting one elementary interaction to the next.

Consequently, every elementary event is organizationally self-contained.

This assumption is not introduced as a physical limitation.

Rather, it reflects the intentionally restricted scope of the present paper.

Persistent organization cannot emerge from a single isolated interaction.

It requires relationships among multiple elementary events, which will be introduced only in the subsequent study of photon formation.

The elementary ψ – θ – χ interaction therefore establishes a memoryless organizational process that serves as the fundamental building block for all higher levels of organization developed throughout the remainder of the framework.

4. Hybrid Organizational Dynamics

The elementary interaction developed in the previous section combines two fundamentally different forms of organizational evolution.

The organizational states ψ and θ evolve continuously throughout the interaction, while χ appears and disappears as a discrete organizational event.

Neither a purely continuous description nor a purely discrete formulation is therefore sufficient.

The elementary ψ – θ – χ interaction must instead be described as a **hybrid organizational dynamical system** in which continuous evolution and discrete transitions coexist within a unified organizational process.

4.1 Continuous Organizational Evolution

Between coupling events, ψ and θ evolve continuously according to their own organizational dynamics.

Their evolution requires no discrete intervention and no organizational restructuring.

The local organizational state changes continuously until the accumulated incompatibility reaches the admissible condition for coupling.

Throughout this stage,

- ψ remains continuous,
- θ remains continuous,
- χ is absent.

The system therefore follows continuous organizational evolution.

No elementary organizational event occurs during this interval.

4.2 Event-Driven Organizational Transitions

Whenever local organizational stress reaches the coupling condition, continuous evolution alone is no longer sufficient.

A discrete organizational event occurs through the appearance of χ .

Unlike ψ and θ , χ is not governed by continuous evolution.

Its existence is entirely event-driven.

Each χ possesses a finite lifetime bounded by two discrete events:

- its appearance during coupling,
- and its disappearance during decoupling.

The elementary interaction therefore alternates naturally between continuous organizational evolution and discrete organizational transitions.

4.3 The Hybrid Dynamical Structure

The coexistence of continuous and discrete dynamics defines the elementary ψ - θ - χ interaction as a hybrid system.

The continuous component consists of the uninterrupted organizational evolution of ψ and θ .

The discrete component consists of the finite existence of χ during temporary organizational restructuring.

The two components are inseparable.

Continuous evolution creates the organizational conditions under which χ appears.

Conversely, the discrete appearance of χ enables the local organizational reconstruction required before continuous evolution can resume.

Neither component can independently describe the complete elementary interaction.

Only their combined operation produces a consistent organizational process.

4.4 Minimal Mathematical Representation

The elementary organizational dynamics may be represented abstractly by distinguishing continuous state evolution from discrete organizational events.

Let

$$\psi(t), \theta(t)$$

denote the continuously evolving organizational states.

The discrete organizational bridge is represented by

$$\chi(t) \in 0,1$$

where

- $(\chi = 0)$ denotes the absence of coupling,
- $(\chi = 1)$ denotes the temporary existence of the elementary organizational bridge.

The elementary dynamics may therefore be expressed conceptually as

$$(\psi, \theta) \rightarrow (\psi, \theta, \chi) \rightarrow (\psi, \theta)$$

where the first transition represents coupling and the second represents decoupling.

This notation is intentionally minimal.

Its purpose is not to specify detailed evolution equations but to distinguish the continuous organizational variables from the discrete organizational event governing elementary interaction.

4.5 Scope of the Hybrid Formulation

The hybrid formulation introduced here describes only the internal organization of one elementary interaction.

No assumptions are made regarding repeated coupling, organizational persistence, cyclic reconstruction, photon formation, or collective propagation.

Those phenomena require relationships among multiple elementary interactions and therefore lie beyond the scope of the present paper.

The role of the hybrid formulation is consequently foundational rather than complete.

It establishes the minimal organizational language required to describe how continuous organizational states and discrete organizational transitions coexist within a single elementary ψ - θ - χ interaction.

Every higher organizational process developed in the subsequent papers will build upon this elementary hybrid architecture without modifying its underlying principles.

5. Scope and Outlook

The formulation developed in this paper intentionally represents only the elementary organizational layer of the ψ - θ - χ framework.

By isolating a single interaction between ψ and θ , the analysis identifies the smallest organizational event capable of producing a transient but internally consistent coupling through the appearance of χ .

This restricted scope is deliberate.

Its purpose is not to simplify the framework artificially but to establish a rigorous organizational foundation before introducing more complex phenomena.

Several important organizational processes are therefore intentionally excluded.

No assumptions have been made regarding repeated interactions, organizational memory, cyclic reconstruction, propagation, photons, collective organization, or electromagnetism.

Each of these phenomena requires relationships among multiple elementary interactions and cannot be derived from a single isolated event alone.

Consequently, the present formulation should not be interpreted as a complete physical model.

Rather, it defines the elementary organizational building block from which higher organizational structures may subsequently emerge.

This distinction is essential.

The elementary interaction introduced here possesses no intrinsic persistence.

Each χ appears only temporarily and disappears once the local organizational reconstruction has been completed.

Likewise, every coupling event is organizationally independent of every other coupling event.

Persistent organization therefore cannot arise within the elementary interaction itself.

If persistent structures exist, they must emerge through new organizational principles governing the relationships among multiple elementary events.

The next paper addresses precisely this question.

Instead of examining one isolated interaction, it investigates how successive elementary $\psi-\theta-\chi$ interactions can become recursively organized into a stable cyclic process.

The central problem is no longer the existence of an elementary interaction but the emergence of organizational persistence from repeated elementary reconstruction.

The elementary interaction developed in the present paper therefore serves as the conceptual and mathematical foundation upon which the remainder of the $\psi-\theta-\chi$ framework is constructed.

Every higher organizational level introduced in the subsequent papers preserves the elementary principles established here while extending them through progressively richer forms of organizational integration.

The analysis presented in this paper thus establishes the first indispensable step toward a fully organizational description of photon formation and, subsequently, collective electromagnetic organization.

Acknowledgments

The author gratefully acknowledges the colleagues whose discussions, questions, and constructive critiques contributed to improving the clarity and consistency of this theoretical framework.

This research was conducted independently and without institutional affiliation or external financial support.

Artificial intelligence tools were used exclusively for technical assistance, including language refinement, structural consistency checks, formatting, and equation verification. All conceptual development, theoretical formulations, mathematical constructions, scientific conclusions, and the overall research direction presented in this work are original contributions of the author.

This paper represents one step within the ongoing development of the **Generative Architecture** framework. Subsequent papers will progressively extend the concepts introduced here into a unified theoretical structure.